

Experiment # 06

Virtual Local Area Networks

A VLAN is a group of end stations in a switched network that is logically segmented by function, project team, or application, without regard to the physical locations of the users. VLANs have the same attributes as physical LANs, but you can group end stations even if they are not physically located on the same LAN segment.

Any port can belong to a VLAN and unicast, broadcast, and multicast packets are forwarded and flooded only to end stations in that VLAN. Each VLAN is considered a logical network. Packets destined for stations that do not belong to the VLAN must be forwarded through a router.

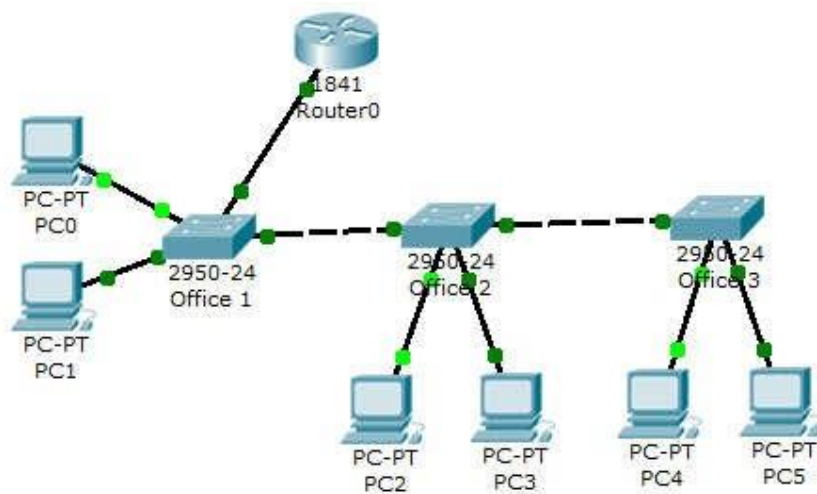
Advantages of VLAN

VLAN provides following advantages:-

- Solve broadcast problem.
- Reduce the size of broadcast domains.
- Allow us to add an additional layer of security.
- Make device management easier.
- Allow us to implement the logical grouping of devices by function instead of location.

LAB Setup

To replicate a given scenario create a topology in packet tracer, as shown in following image.



Configurations used in this topology are following:

PC's Configurations

Device	IP Address	Subnet Mask	Gateway	VLAN	Connected With
PC0	10.0.0.2	255.0.0.0	10.0.0.1	VLAN 10	Office 1 Switch on F0/1
PC1	20.0.0.2	255.0.0.0	20.0.0.1	VLAN 20	Office 1 Switch on F0/2
PC2	10.0.0.3	255.0.0.0	10.0.0.1	VLAN 10	Office 2 Switch on F0/1
PC3	20.0.0.3	255.0.0.0	20.0.0.1	VLAN 20	Office 2 Switch on F0/2
PC4	10.0.0.4	255.0.0.0	10.0.0.1	VLAN 10	Office 3 Switch on F0/1
PC5	20.0.0.4	255.0.0.0	20.0.0.1	VLAN 20	Office 3 Switch on F0/2

Office 1 Switch Configuration

Port	Connected to	VLAN	Link	Status
F0/1	With PC0	VLAN 10	Access	OK
F0/2	With PC1	VLAN 20	Access	OK
F0/24	With Switch2	VLAN 10,20	Trunk	STP - Blocked
F0/3	With Router	VLAN 10,20	Trunk	STP – Blocked

Office 2 Switch Configuration

Port	Connected to	VLAN	Link	Status
F0/1	With PC0	VLAN 10	Access	OK
F0/2	With PC1	VLAN 20	Access	OK
F0/24	With Switch1	VLAN 10,20	Trunk	STP - Blocked
F0/23	With Switch3	VLAN 10,20	Trunk	STP – Blocked

Office 3 Switch Configuration

Port	Connected to	VLAN	Link	Status
F0/1	With PC0	VLAN 10	Access	OK
F0/2	With PC1	VLAN 20	Access	OK
F0/24	With Switch2	VLAN 10,20	Trunk	STP – Blocked

Step 1: Configure VTP Server and Client in Switch

VLAN Trunk Protocol (VTP) is a Cisco proprietary protocol used to share VLAN configuration across the network. Main goal of VTP is to manage all configured VLANs across the network.

In our network Office1 Switch is configured as VTP (VLAN Trunk Protocol) Server. The Office2 and Office3 switches are configured as VTP clients. We only need to create VLANs in VTP Server. VTP Server will propagate this information to all VTP clients.

By default all switches work as VTP server so we only need few commands to configure it. In following commands we will

- Set hostname to **S1, S2 and S3** respectively.
- Set domain name to *LabTask*
- Set password to 12345. (Password is case sensitive)

Access CLI prompt of **Office 1 Switch** and execute following commands.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#vtp mode server
Device mode already VTP SERVER.
S1(config)#vtp domain LabTask
Changing VTP domain name from NULL to LabTask
S1(config)#vtp password 12345
Setting device VLAN database password to 12345
```

We will configure Office 2 Switch and Office 3 Switch as VTP client switch.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S2
S2(config)#vtp mode client
Setting device to VTP CLIENT mode.
S2(config)#vtp domain LabTask
Changing VTP domain name from NULL to LabTask
S2(config)#vtp password 12345
Setting device VLAN database password to 12345
```

Now access **CLI** prompts of **Office 3 Switch** and enter following commands

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S3
S3(config)#vtp mode client
Setting device to VTP CLIENT mode.
S3(config)#vtp domain LabTask
Changing VTP domain name from NULL to LabTask
S3(config)#vtp password 12345
Setting device VLAN database password to 12345
```

We have configured VTP server and VTP client. At this moment VTP client will not receive VTP messages from server. We need to configure DTP (Dynamic Trunking Protocol) between switches. We will configure DTP along with VLAN Membership commands. For your convenience just note that the DTP commands are like:

```
Switch(config)#interface fastEthernet 0/24
Switch (config-if)#switchport mode trunk
Switch(config-if)#exit
```

Step 2: How to create VLAN *vlan vlan-number*

command is used to create the VLAN.

Office 1 Switch

```
S1(config)#vlan 10
S1(config)#name Sales
S1(config-vlan)#exit
```

```
S1(config)#vlan 20
S1(config)#name Management
S1(config-vlan)#exit
S1(config)#
```

➤ Assign VLAN Membership

switchport access vlan [vlan number] command is used to assign VLAN to the interface. Following commands will assign VLANs to the interfaces.

Office 1 Switch

```
S1(config)#interface fastEthernet 0/1
S1(config-if)#switchport access vlan 10
S1(config-if)#interface fastEthernet 0/2
S1(config-if)#switchport access vlan 20
S1(config)#interface fastEthernet 0/3
S1(config-if)#switchport mode trunk
S1(config)#interface fastEthernet 0/24
S1(config-if)#switchport mode trunk
S1(config-if)#exit
```

Office 2 Switch

```
S2(config)#interface fastEthernet 0/1
S2(config-if)#switchport access vlan 10
S2(config-if)#interface fastEthernet 0/2
S2(config-if)#switchport access vlan 20
S2(config)#interface fastEthernet 0/23
S2(config-if)#switchport mode trunk
S2(config)#interface fastEthernet 0/24
S2(config-if)#switchport mode trunk
S2(config-if)#exit
```

Office 3 Switch

```
S3(config)#interface fastEthernet 0/1
S3(config-if)#switchport access vlan 10
S3(config-if)#interface fastEthernet 0/2
S3(config-if)#switchport access vlan 20
S3(config)#interface fastEthernet 0/24
S3(config-if)#switchport mode trunk
S3(config-if)#exit
```

We have successfully assigned VLAN membership and DTP configuration. It's time to test our configuration. To test this configuration, we will use *ping* command. *ping* command is used to test connectivity between two devices. As per our configuration, devices from same VLAN can communicate. Devices from different VLANs must not be able to communicate with each other without router.

Test VLAN configuration

Ping any PC from any other but belonging to the same VLAN. For example: ping PC 10.0.0.3 from PC 10.0.0.2

Step 3: Configure Router on Stick

Typically routers are configured to receive data on one physical interface and forward that data from another physical interface based on its configuration. Each VLAN has a layer 3 address that should be configured as default gateway address on all its devices. In our scenario we reserved IP address 10.0.0.1 for VLAN 10 and 20.0.0.1 for VLAN 20.

With default configuration we need two physical interfaces on router to make this inter VLAN communication. Due to price of router, it's not a cost effective solution to use a physical interface of router for each VLAN. Usually a router has one or two Ethernet interface. For example if we have 50 VLANs, we would need nearly 25 routers in order to make inter VLANs communications. To deal with situation we use Router on Stick.

Router on Stick is router that supports trunk connection and has an ability to switch frames between the VLANs on this trunk connection. On this router, single physical interface is sufficient to make communication between our both VLANs.

Access command prompt of Router Run following commands in same sequence to configure Router on Stick.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastEthernet 0/0
Router(config-if)#no ip address
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface fastEthernet 0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 10.0.0.1 255.0.0.0
Router(config-subif)#exit
Router(config)#interface fastEthernet 0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 20.0.0.1 255.0.0.0
Router(config-subif)#exit
```

```
Router#vlan database
Router(vlan)#vlan 10 name sales
VLAN 10 added:
Name: sales
Router(vlan)#vlan 20 name management
VLAN 20 added:
Name: management
Router(vlan)#exit
APPLY completed. Exiting....
```

- In above configuration we broke up single physical interface [FastEthernet 0/0] into two logical interfaces, known as sub-interfaces. Router supports up to 1000 interfaces including both physical and logical.
- By default interface link works as access link. We need to change it into trunk link. encapsulation commands specify the trunk type and associate VLAN with sub-interface.
- In next step we assigned IP address to our sub-interface.

That's all configuration we need to switch VLANs. Now we can test different VLAN communications. To test inter VLANs communication open command prompt of PC and ping the PC of other VLAN.

PC [10.0.0.3] from VLAN 10 can now access PC [20.0.0.2] from VLAN 20.

To see the VLAN configurations following command is used

Switch# show vlan brief